AGA KHAN UNIVERSITY EXAMINATION BOARD

SECONDARY SCHOOL CERTIFICATE

CLASS IX EXAMINATION

MAY 2015

Physics Paper II

Time: 2 hours 25 minutes Marks: 40

INSTRUCTIONS

Please read the following instructions carefully.

1. Check your name and school information. Sign if it is accurate.

I agree that this is my name and school. **Candidate's signature**

2. RUBRIC. There are ELEVEN questions. Answer ALL questions. Questions 9, 10 & 11 each offer TWO choices. Attempt any ONE choice from each.

3. When answering the questions:

   Read each question carefully.
   Use a black pencil for diagrams. DO NOT use coloured pencils.
   DO NOT use staples, paper clips, glue, correcting fluid or ink erasers.
   Complete your answer in the allocated space only. DO NOT write outside the answer box.

4. The marks for the questions are shown in brackets ( ).

5. You may use a simple calculator if you wish.

Q.1. (Total 3 Marks)

'Physics helps in the development of our society.' In the light of the given statement mention any THREE important roles of physics giving a brief description of each.

Q.2. (Total 3 Marks)

Give any ONE difference between each of the following pairs.

Distance	Displacement

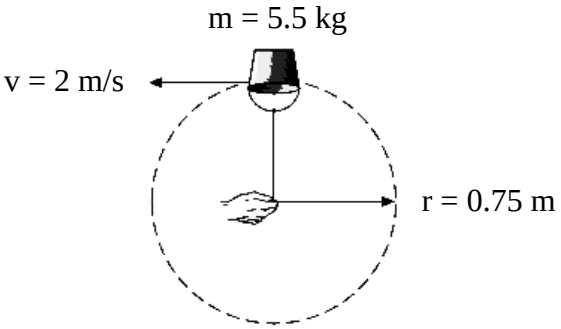
Speed	Velocity

Scalar Quantities	Vector Quantities

Q.3. (Total 3 Marks)

In the given diagram, calculate the centripetal force acting on the bucket of water.

(Note: Extract data from the given diagram and provide your answer in SI unit.)



Q.4. (Total 3 Marks)

A body that remains at rest or moves uniformly even though a number of forces are acting on it is said to be in equilibrium. There are three states of equilibrium.

Write any ONE example of each state of equilibrium from daily life. For each example, make clear which state of equilibrium is involved.

Q.5. (Total 3 Marks)

How does the gravitational force cause artificial satellites to orbit the Earth?

_____

_____

_____

_____

_____

_____

Q.6. (Total 3 Marks)

An object falls from the height of 40 m. Calculate its velocity, when it strikes the ground where the value of 'g' = 9.8 m/s^2.

(Note: Extract data from the question and provide your answer in SI unit.)

Q.7. (Total 3 Marks)

Snowshoes are flat frames that could be attached to the bottom of shoes, so that a skier can walk on the deep snow without sinking.

Describe using THREE points how the snowshoes prevent a skier from sinking into the soft snow.



Snowshoe



Normal shoe

_____

_____

_____

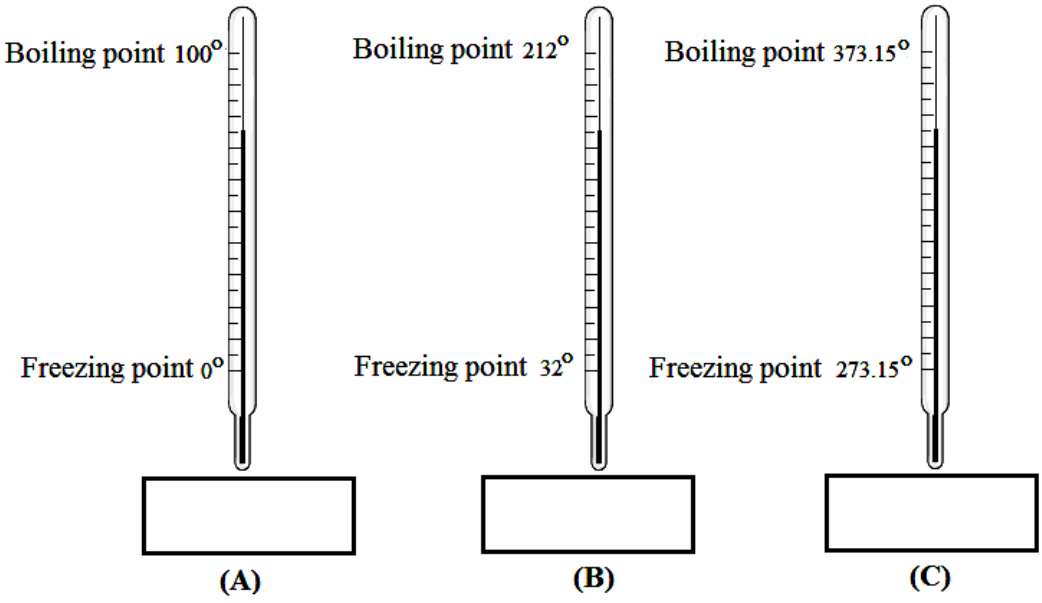
_____

_____

_____

Q.8. (Total 4 Marks)

a. Identify the temperature scales of the following thermometers and write the scales in the boxes. (3 Marks)



b. The reading of the thermometer (A) is approximately _____. (1 Mark)

Q.9. (Total 5 Marks)

EITHER

a. A person starts driving from his home to the petrol pump which is 1400 m away. After travelling 1600 m, he realizes that he has passed the petrol pump. He then turns his car and drives back to the pump and completes his journey to the pump in a total of 5 minutes.

Determine the total distance covered and the average speed of the car during his entire journey.

(Note: Extract data from the question and provide your answers in SI units.)

OR

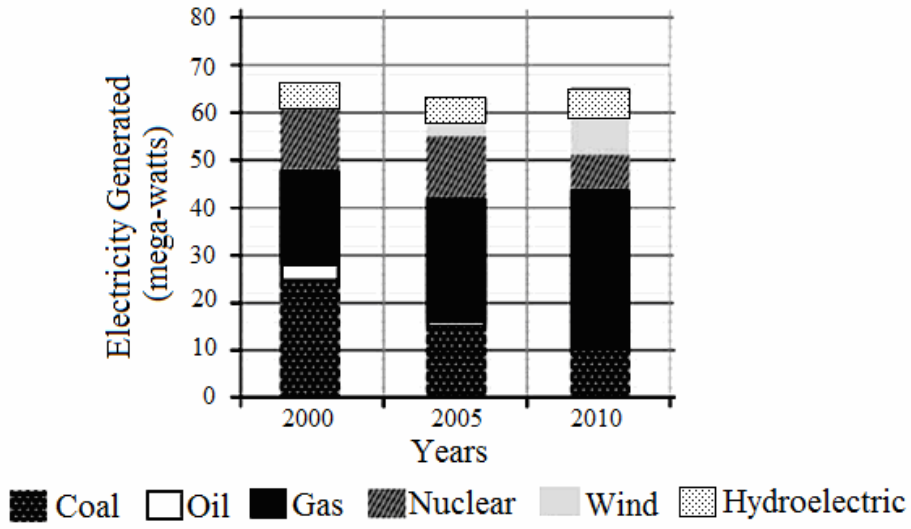
b. A stationary body acquires a velocity of 20 m/s after a force of 100 N acts on it for 5 s. Find the mass and the momentum of that body at 5 s.

(Note: Extract data from the question and provide your answers in SI units.)

Q.10. (Total 5 Marks)

EITHER

a. The given graph shows the energy sources used by a small scale electricity generation plant from the year 2000 to 2010. In this graph, the electricity produced in mega-watt units is plotted on the *y*-axis while the time in years is on the *x*-axis.



i. Name any ONE renewable energy source used for power generation shown in the graph.

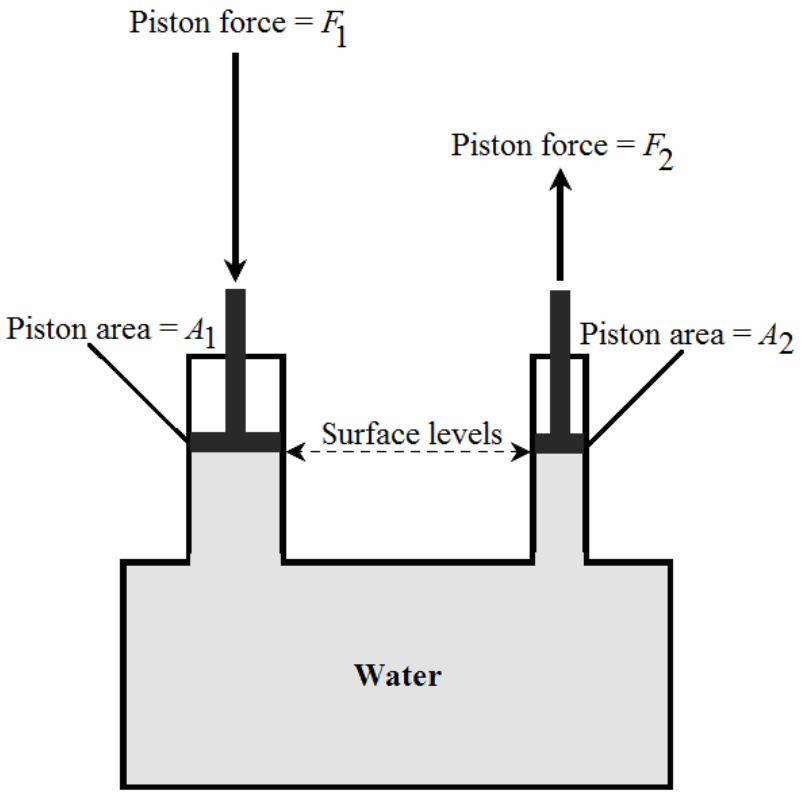
ii. Approximately how many mega-watts of electricity is generated from non-renewable energy sources in the year 2005?

iii. Name the largest power generation source in the year 2010.

iv. Highlight any TWO environmental issues associated with non-renewable energy sources.

OR

b. In the illustrated system, if the piston with the area A_1 is pushed downwards with a force F_1, using Pascal's law, explain the upward force F_2 on the piston with the area A_2.



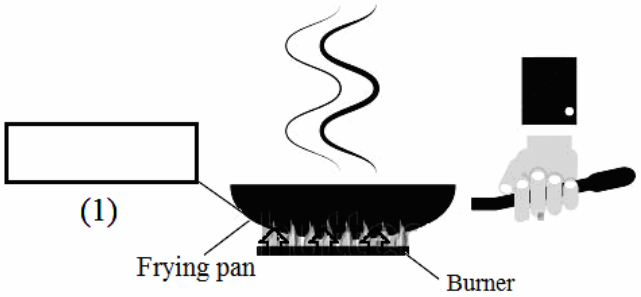
Q.11. (Total 5 Marks)

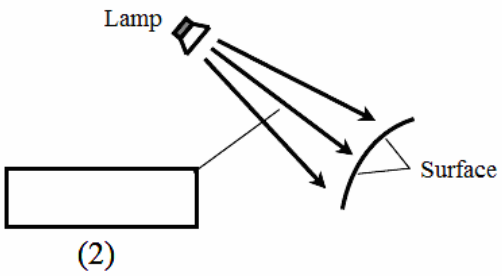
EITHER

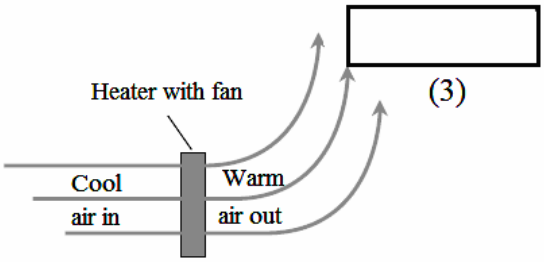
a. Evaporation occurs at all temperatures whereas boiling takes place at a specific temperature which varies depending on the external pressure. Justify the given statement.

OR

b. Name the type of heat transfer in the given boxes and explain any ONE of them.







END OF PAPER

Please use this page for rough work